

Probability and Statistics

Lecture 1:

- * Collection
- * Interpret
- * Classification
- * Analysis

Descriptive stats

collection and analysis of data

Inferential Stats

Drawing conclusion from sample data.

- * Population : objects having common characteristics.
size of population is denoted by (N) .
- * sample : smallest part of population. size of sample is denoted by (n) .

Parameter

- numerical value calculated from population.
- fixed numbers
- denoted by greek letters
- mean denoted by μ
- std. deviation denoted by σ

Statistic

- numerical value calculated from statistic
- variable values
- denoted by english, roman, italic letters
- mean denoted by $\bar{x}$
- std. deviation denoted by s

Lecture 2:

Variables

changes with time, space

Quantitative

can be measured

Discrete

always in whole no.

Continuous

can be in points

Qualitative

cannot be measured

Primary Data

data that is collected directly and not has undergone any statistical treatment.

- * direct personal investigation (interviews)
- * indirect investigation (baaton baaton mei puchna)
- * collection through questionnaire
- * collection through enumerators (train people to collect information for you)
- * collection through local source

Secondary Data

data that has undergone some statistical change.

- * officials (people from govt)
- * semi-officials (people from semi-official company)
- * Journals and Newspaper
- * Research and organizations

Lecture 3:

One-way Classification

when data is classified based on one variable

Two-way Classification

when data is classified based on two variables

Multi-way classification

when data is classified based on more than two variables.

Clockwise

Define data starting from main to sub characteristics

Anti-clockwise

Define data starting from sub to main characteristics.

Frequency

No. of observations in a class. It is denoted by "f"

Frequency Distribution

a table which has different classes and their respective frequencies against them is called frequency distribution or frequency distribution table or group frequency distribution.

Class Limits (C.L)

when we have two consecutive classes and the max value of first class and min. value of second class are not same then those classes are called class limits. Both lower & upper value of class limit are inclusive in the same class.

* Lower class Limit

minimum values of class limits.

* Upper class Limit

maximum values of class limits

Class Boundaries (C.B)

when we have two consecutive classes and the max value of first class and min value of second class are same then those classes are called class boundaries.

* Lower class Boundary

minimum values of class boundaries

* Upper class Boundary

maximum values of class boundaries

Both upper & lower value of class boundary are not inclusive in the same class. The upper is exclusive.

Conversion from class Limit to Class Boundary

7 — 13	when difference is same in all $\frac{13-14}{2} = \frac{1}{2} = 0.5$
14 — 20	
21 — 27	
28 — 34	
So,	subtract the result from all the lower class limits and add the result from to all the upper class limits.
6.5 — 13.5	
13.5 — 20.5	

8 — 14	when difference is not same in all then add upper class limit of 1st class and lower class limit of 2nd class and divide by 2. Replace those limits with the result.
15 — 20	
22 — 30	
So,	

$$7.5 - 14.5$$

$$14.5 - 21$$

$$21 - 31$$

Class Interval

class interval / height / width is difference between the upper and lower class limits or upper or lower class boundaries.

In case of class boundaries, if we subtract lower class boundary from upper class boundary then it also gives class interval.

class interval is denoted "h".

Class Mark

when we add the lower ~~for~~ and upper limits of same class and divide them by 2 then we get the midpoint / class mark. It is denoted by X .

Ex :		X	class interval	$10 - 3 = 7$
	3 - 9	6		
	10 - 16	13		
	17 - 23	20		
	24 - 30	27		
	31 - 37	34		
	38 - 44	41		

Lecture 4:

Relative Frequency

% contribution of each class in the frequency distribution. It is denoted by rf / RF . It is ^{also} called "probability".

$$\frac{\text{frequency of 1 class}}{\text{sum of total frequencies}}$$

Commulative Frequency

adding frequency of first class with second class. Its denoted by CF .

Commulative Relative Frequency

e.g.	f	CF	crf
	8	8	$8/36 = 0.2$
	12	20	$20/36 = 0.5$
	10	30	$30/36 = 0.8$
	6	36	$36/36 = 1$

Construction of Grouped Frequency Distribution

- i- Decide the no. of class into which the data to be grouped
- ii- Find range of data
- iii- Divide range by no. of classes
- iv- Decide where to locate lower class limit
- v- Determine the remaining class limits
- vi- Divide data into classes
- vii- Total the frequency column.

Ex : range : 200 - 600

observations

210	340	600	414	438
250	420	318	508	505
240	510	514	515	216
360	520	508	206	304
410	509	360	218	513
500	350	342	513	580
599	428	532	308	384
600	503	405	384	540

No. of classes

$$\begin{aligned}K &= 1 + 3.3 \log N \\&= 1 + 3.3 \log(40) \\K &= 6.287\end{aligned}$$

N = no. of observations

if ans is less than 0.5 then floor func otherwise (≥ 0.5) ceil func.
So, No. of classes are 6.

Range

$$R = \text{max value} - \text{min value}$$

$$= 600 - 206$$

$$R = 394$$

class interval

$$h = \frac{R}{K} = \frac{394}{6}$$

$$h = 65.66$$

$$h = 66$$

Deciding Classes Limits

206 — 272

272 — 337

338 — 403

404 — 469

470 — 535

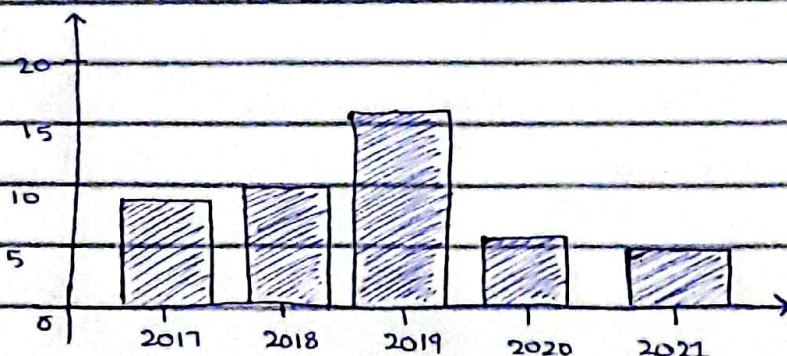
536 — 601

min limit + h

Q# The following data is about no. of students admitted to fast uni during last five years.

Years	no. of std (variable)
2017	8
2018	10
2019	16
2020	6
2021	5

we will use bar chart..
one variable.



Title, x-axis, y-axis,
source

Lecture 5:

In case of **Simple Bar chart**,

If data is not dependant on time then choose ascending ~~and~~ descending order for making the graph.

Multi-bar chart

Q# The following data is about the budget allocation of two different organizations. In its different depts.

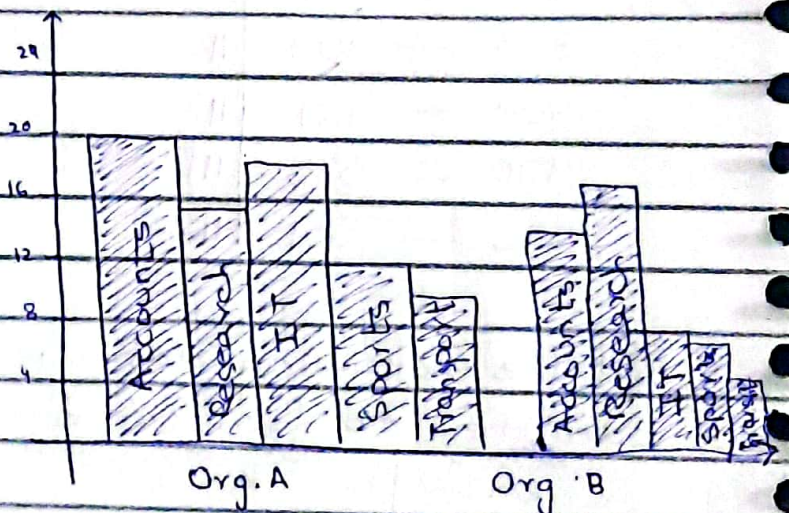
Depts.	Org. A (L)	Org. B (R)
Accounts	20	13
Research	15	17
IT	18	08
Sports	12	07
Transport	10	05
	= 15	= 50

if we make simple bar chart then we will have to make two, one for each org.

sequence must be same for

Both organizations

Title, x-axis, y-axis, source

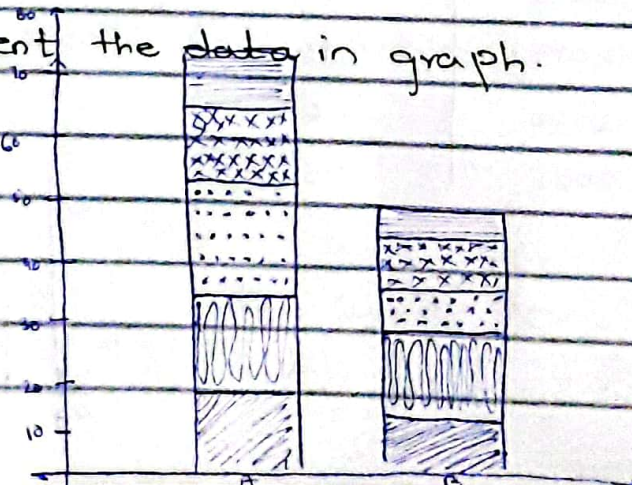


Component Bar chart

Q# Based on previous, present the data in graph.

Title, x-axis, y-axis,

source



Rectangles and Subdivided Rectangles

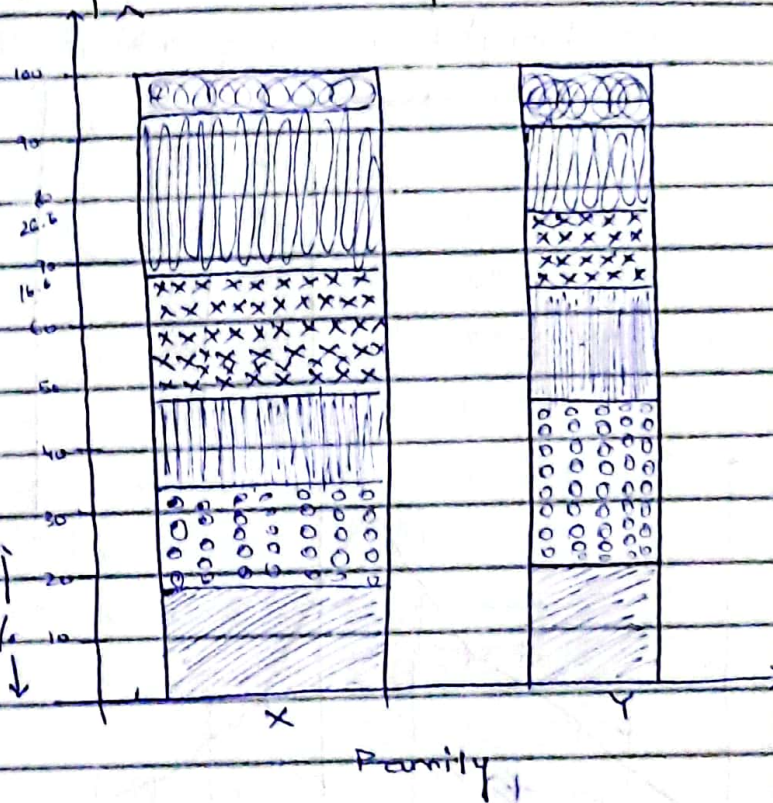
When we have to compare data.

Q# The following info is about the monthly expenditure of two different families. Compare their expenditure with help of chart.

Items	Fam X	Fam Y
Food	18 ^{16%}	12 ^{20%}
Clothing	16 ^{16%}	16 ^{27%}
House Rent	14 ^{14%}	10 ^{17%}
Fuel & Light	20 ^{20%}	8 ^{13%}
Education	26 ^{26%}	4 ^{6%}
Misc	6 ^{6%}	10 ^{17%}
	<u>100</u>	<u>60</u>

The width of the bar depends on grand total of the both families

Title, x-axis, y-axis, source



Pie / Sector Chart

$$\text{angle} = \frac{\text{component part}}{\text{whole part}} \times 360^\circ$$

Fam X

64.8

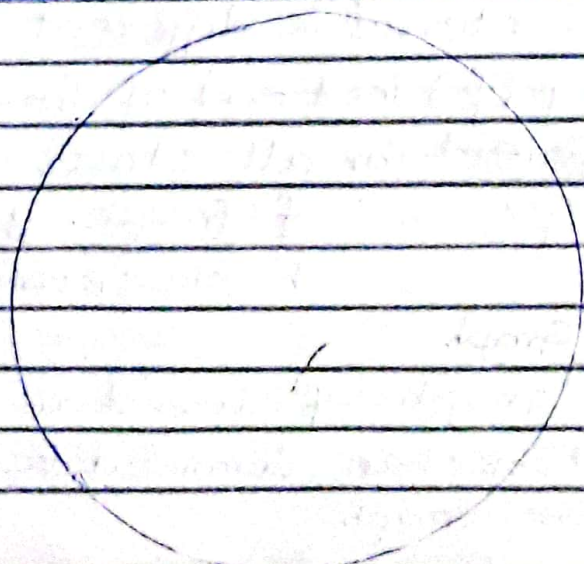
57.6

50.4

72

93.6

21.6



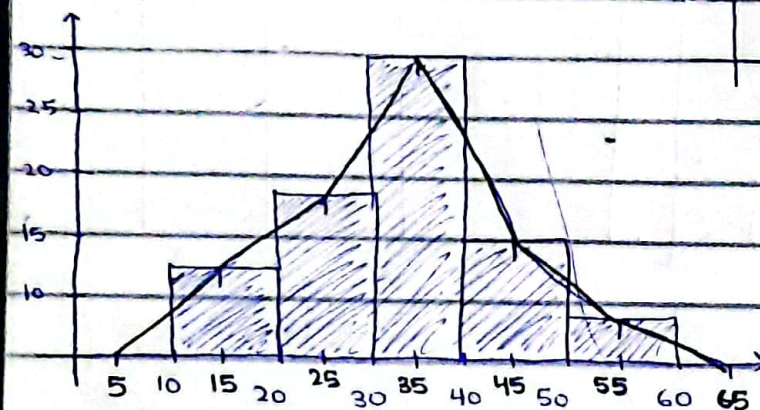
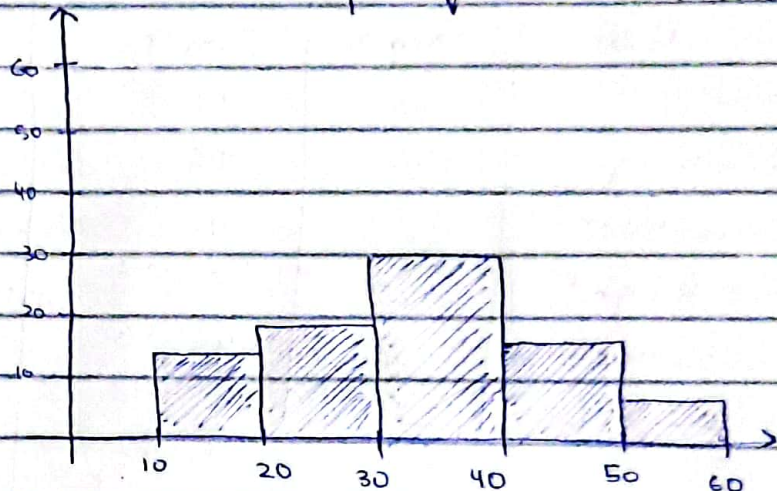
Lecture 6:

Histograms

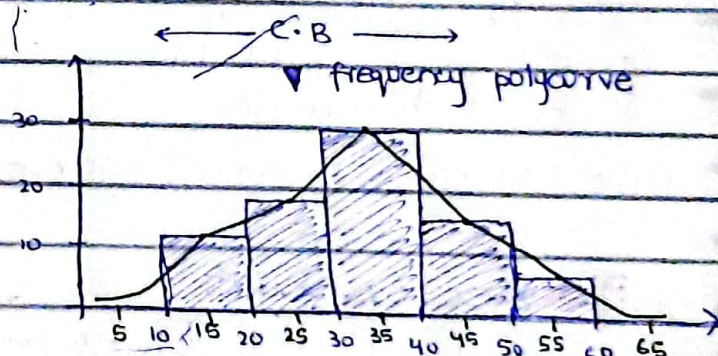
It is set of adjacent rectangles whose bases are marked by class boundaries and not class limits.

The height is proportional to the frequencies.

Class Boundaries	f	x
10 — 20	12	15
20 — 30	18	25
30 — 40	30	35
40 — 50	15	45
50 — 60	8	55



▲ frequency polygon (trend of data)



- * If class interval is different then we plot frequency density (f') instead of frequency and width of bar is different for all classes.

$$f' = \frac{f(\text{frequency})}{h(\text{class interval})}$$

Ogive Graph

- * The graph of Less than Cumulative Frequency or More than Cumulative Frequency is called Ogive Graph

Note:

- * If we are given only mid-point (x) and frequency and we need to construct histogram. If data is discrete then we can't construct histogram. ^{otherwise} find difference between two mid-points and ^{divide by 2} for upper class limits, add the ans to all mid-points and subtract for lower class boundaries.

Central Tendency

The tendency of observations lying in specific part ~~of data~~ of data.

The methods or techniques to find central part of data is called measures of central Tendencies.

1) Arithmetic mean / mean / Average / A.M

(a) Weighted Arithmetic Mean

2) Median

(a) Quartiles

(b) Deciles

(c) Percentiles

3) Mode

4) Geometric Mean / G.M

5) Harmonic Mean / H.M

Positional
Averages

Mathematical
Averages

Arithmetic Mean

$$A.M = \frac{\text{sum of all observations}}{\text{Total no. of observations}}$$

$$H = \frac{x_1 + x_2 + \dots + x_N}{N}$$

$N \rightarrow$ population

$$\mu = \frac{\sum X}{N} \quad \bar{X} = \frac{\sum x}{n}$$

(Population)

(Sample)

$$\mu = \frac{\sum fX}{\sum f}$$

$$\bar{X} = \frac{\sum fx}{\sum f}$$

Grouped Data

Lecture 7:

Ex : X

20

24

18

26

25

23

26

if not specified that data sample / population then consider it sample.

$$\bar{X} = \frac{\sum x}{n} = \frac{162}{7} = 23.14$$

$$\Rightarrow \sum X = n \times \bar{X}$$

$$\Rightarrow n = \frac{\sum X}{\bar{X}}$$

≈ 23 (data is not continuous)

Ex : class limit

f

X

fX

20 - 28

12

24

288

28 - 34

16

41

656

54 - 60

10

57

570

60 - 78

30

69

2070

78 - 98

12

88

1056

$$\bar{X} = \frac{\sum fx}{\sum f} = \frac{4640}{80}$$

$$\boxed{\bar{X} = 58}$$

$$\Rightarrow \sum f = \frac{\sum fx}{\bar{X}}$$

$$\sum fx = 4640$$

$$\Rightarrow \sum fx = \bar{X} \cdot \sum f$$

Properties of Arithmetic Mean

- * Sum of deviation of all the observations from its mean is always equal to zero.

$$\sum (x - \bar{x}) = 0$$

- * Sum of square deviations of all the observations from its mean is always minimum.

$$\sum (x - \bar{x})^2 \leq \sum (x - A)^2 \quad \text{where } A \text{ is any no.}$$

- * If we have k subgroups with their sizes $n_1, n_2, n_3, \dots, n_k$ along with their means $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k$ then the combine mean for the whole data is

$$\bar{X}_c = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + \dots + n_k \bar{x}_k}{n_1 + n_2 + n_3 + \dots + n_k}$$

$$\bar{X}_c = \frac{\sum n \bar{x}}{\sum n}$$

Q # $n_1 = 20, \bar{x}_1 = 30, n_2 = ?, \bar{x}_2 = 26, \bar{X}_c = 58$

$$\bar{X}_c = 58 = \frac{20(30) + 26n_2}{20 + n_2}$$

$$20 + n_2 = \frac{20(30)}{58} + \frac{26n_2}{58}$$

$$20 + n_2 = \frac{300}{29} + \frac{13n_2}{29}$$

$$580 + 29n_2 = 300 + 13n_2$$

$$29n_2 - 13n_2 = 300 - 580$$

$$16n_2 = -280$$

$$n_2 = -17.5$$

$$\Rightarrow |n_2 = 18|$$

Weighted Arithmetic Mean

A.M is calculated when all observations are given equal importance. But there are some cases in which different observations may get different importance. In such cases, we assign them weights.

If we have k observations $x_1, x_2, \dots, x_k$ having weights $w_1, w_2, \dots, w_k$ then the weighted mean can be calculated as

$$\bar{X}_w = \frac{x_1 w_1 + x_2 w_2 + \dots + x_k w_k}{w_1 + w_2 + \dots + w_k}$$

Mode

The most repeating value in the data.
It is $\hat{X}$.

→ Uni Modal Distribution

when only one mode exists in the data.

→ Bi Modal Distribution

when only two mode exists in the data.

→ Multi Modal Distribution

when more than two mode exists in the data.

It's calculated by,

$$\hat{X} = lb + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \times h$$

f_m → maximum frequency to decide class model

lb → Lower class boundary of the modal class

h → class height of modal class

f_1 → the frequency of class above modal class. If not then 0

f_2 → the frequency of class below modal class. If not then 0.

Lecture 8:

Median

It is a point which divides the data into two equal parts. Each part contains 50% observations of the data. One part having 50% observations above and other having 50% below, when data is arranged in ascending or descending order. (only in case of ungrouped data, the data is sorted in ascending / descending)

It is denoted by " $\tilde{X}$ ".

$$\tilde{X} = \left(\frac{n+1}{2} \right)^{\text{th}}$$

→ ungrouped data

n : no. of observations

Ex: 52, 75, 16, 35, 78, 93, 25, 10, 93, 82, 70, 48, 51, 72, 6

arranging ⇒

6, 10, 16, 25, 35, 48, 51, 52, 70, 72, 75, 78, 82, 93, 93

$$\tilde{X} = \frac{15+1}{2} = 8^{\text{th}} \text{ position}$$

So, Median = 52

Note :

if position is in points, then add the ^{data of} previous position and ^{data of} next position and divide by 2 which will give median.

$$\tilde{X} = lb + \frac{h}{f} \left(\frac{\sum f}{2} - c \right) \quad \rightarrow \text{grouped data}$$

lb : lower boundary of median class

h : class interval of median class

f : frequency of median class

c: cumulative frequency preceding to the median class

Note:

Solve $\Sigma f/2$ and check its ans. in CF column to decide median class.

Ex:	Class Limit	f	CF
	12 — 30	7	7
	31 — 39	28	35
61.5 →	40 — 56	35	70
	57 — 71	16	86
	72 — 94	12	98
	95 — 103	20	118
	104 — 110	5	123

$$\frac{\Sigma f}{2} = \frac{123}{2} = 61.5$$

⇒ 40 — 56 is the median class.

change into class boundaries

$$[39.5 - 56.5]$$

$$h = 56.5 - 39.5$$

$$[h = 17]$$

$$[f = 35]$$

$$[lb = 39.5]$$

$$[C = 35]$$

So,

$$\tilde{X} = 39.5 + \frac{17}{35} (61.5 - 35)$$

$$\tilde{X} = 52.37$$

Quartiles

These are the values which divides the data in 4 equal parts. Each part will contain 25% observations of the data. They are denoted by Q_1, Q_2, Q_3 .

Q_1 → lower quartile. It has 25% observations below it and 75% observations above it.

Q_2 → median quartile. It has 50% observations below it and 50% above it.

Q_3 → Upper quartile. It has 75% observations below it

and 25% above it.

Ungrouped Data

$$Q_1 = \left(\frac{n+1}{4} \right)^{\text{th}}$$

$$Q_2 = 2 \left(\frac{n+1}{4} \right)^{\text{th}}$$

$$Q_3 = 3 \left(\frac{n+1}{4} \right)^{\text{th}}$$

Grouped Data

$$Q_1 = lb + \frac{h}{f} \left(\frac{\sum f}{4} - c \right)$$

$$Q_2 = lb + \frac{h}{f} \left(\frac{2 \sum f}{4} - c \right)$$

$$Q_3 = lb + \frac{h}{f} \left(\frac{3 \sum f}{4} - c \right)$$

Deciles

They divide the data into 10 equal parts. It is denoted by " $D_1, D_2, D_3, D_4, \dots, D_9$ ".

Ungrouped Data

$$D_1 = \left(\frac{n+1}{10} \right)^{\text{th}}$$

$$D_2 = 2 \left(\frac{n+1}{10} \right)^{\text{th}}$$

⋮

$$D_9 = 9 \left(\frac{n+1}{10} \right)^{\text{th}}$$

Grouped Data

$$D_1 = lb + \frac{h}{f} \left(\frac{\sum f}{10} - c \right)$$

$$D_2 = lb + \frac{h}{f} \left(\frac{2 \sum f}{10} - c \right)$$

⋮

$$D_9 = lb + \frac{h}{f} \left(\frac{9 \sum f}{10} - c \right)$$

Percentiles

They divide data into 100 equal parts. It is denoted by " $P_1, P_2, P_3, P_4, \dots, P_{99}$ ".

Ungrouped Data

$$P_1 = \left(\frac{n+1}{100} \right)^{\text{th}}$$

Grouped Data

$$P_1 = lb + \frac{h}{f} \left(\frac{\sum f}{100} - c \right)$$

$$P_{99} = 99 \left(\frac{n+1}{100} \right)^{th}$$

$$P_{99} = lb + h \left(\frac{99 \leq f}{f} - c \right)$$

Ex: Use previous data.

$$\hat{X}_{GD} = 39.5 + \frac{35-28}{(35-28) + (35-16)} \times (56.5 - 39.5)$$

$$\hat{X}_{GD} = 44.08$$

$$\hat{X}_{UGD} = 93$$

$$Q_{3_{ung}} = 3 \left(\frac{15+1}{4} \right) = 12^{th} \Rightarrow Q_{3_{ung}} = 78$$

$$Q_{3_{GD}} = 71.5 + \frac{(94.5 - 71.5)}{12} \times (92.25 - 86)$$

$$Q_{3_{GD}} = 83.47$$

$$D_{4_{UGD}} = 4 \left(\frac{15+1}{10} \right) = 6.4^{th}$$

$$\Rightarrow 6^{th} \text{ val} + (0.4)(7^{th} \text{ val} - 6^{th} \text{ val}) = 48 + 0.4(51 - 48)$$

$$D_{4_{UGD}} = 49.20$$

$$D_{4_{GD}} = 39.5 + \frac{17}{35} (49.2 - 35)$$

$$D_{4_{GD}} = 46.39$$

$$P_{82_{UGD}} = \frac{82(15+1)}{100} = 13.12 \Rightarrow 13^{th} + (0.12)(14^{th} - 13^{th})$$

$$P_{82 \text{ ugn}} = 83.43$$

$$P_{82 \text{ GD}} = 94.5 + \frac{(103.5 - 94.5)}{20} \times (100.86 - 98)$$

$$P_{82 \text{ GD}} = 95.79$$

Note :

- * If data is discrete and grouped then mode is the data having max frequency.
- * If data is discrete and grouped then the decile is the data ^{CF} whose contains the ans of $\left(7. \frac{\sum f + 1}{10}\right)$

Lecture 9:

↑

D₇

Geometric Mean

sudden increase or decrease

It is the n^{th} +ve root of the product of non-zero and non-negative values.

- * $GM = (X_1 \cdot X_2)^{1/2}$ for X_1, X_2
- * $GM = (X_1 \cdot X_2 \cdot X_3)^{1/3}$ for X_1, X_2, X_3
- * $GM = (X_1 \cdot X_2 \dots X_n)^{1/n}$ for n observations

} Ungrouped Data.

or

$$* GM = \text{Antilog} \left(\frac{\sum \log X}{n} \right)$$

} Ungrouped Data

$$* GM = (X_1 f_1 \cdot X_2 f_2 \dots X_n f_n)^{1/\sum f}$$

or

$$* GM = \text{Antilog} \left(\frac{\sum f \log X}{\sum f} \right)$$

} Grouped Data

↓
X = midpoints

Ex: classes	f	X	f log X
12 — 30	7	21	9.25
31 — 39	28	35	43.23
40 — 56	35	48	58.84
57 — 71	16	64	28.89
72 — 94	12	83	23.02
95 — 103	20	99	39.91
104 — 110	5	107	10.14
			213.3

$$GM = \text{Antilog } 213.3$$

123

$$GM = 54.2$$

Harmonic Mean

It is the reciprocal of A.M of the reciprocal of non-zero values.

$$H.M = \frac{n}{\sum (1/x)} \quad \left. \begin{array}{l} \text{UGD} \\ x \rightarrow \text{observation} \end{array} \right\}$$

$$H.M = \frac{\sum f}{\sum (f/x)} \quad \left. \begin{array}{l} \text{GD} \\ x \rightarrow \text{midpoint} \end{array} \right\}$$

Ex: 30, 25, 16

$$\sum \left(\frac{1}{x} \right) = 0.136$$

$$HM = \frac{3}{0.136} = 22.09$$

20, 16, -12

$$\sum \left(\frac{1}{x} \right) = 0.029$$

$$HM = \frac{3}{0.029} \Rightarrow 102.857$$

Note :

* $AM \geq GM \geq HM$

* $GM = \sqrt{(AM)(HM)}$

Measures of Dispersion

Scatterness of all the observations from some central part of data.

Absolute Measure of Dispersion

It is ^{the} one which measure the dispersion in the terms of same unit or in square unit as the unit of the measurement of data.

Relative Measure of Dispersion

It is the one which measure the dispersion in the terms of ratios / coefficients / percentage. It is unit free quantity. and useful for comparison of datasets with different unit of measurements.

Lecture 10:

Absolute MOD	Relative MOD
(1) Range	Coefficient of range
(2) Quartile deviation	Coefficient of quartile deviation
(3) Mean deviation	Coefficient of mean deviation
(4) Variance	} Coefficient of variation
(5) Standard deviation	

Range

It is the difference b/w max and min observation

of data, simplest & poorest measure of dispersion.

$$R = X_{\max} - X_{\min} \quad \text{] Ungrouped}$$

$$R = \text{Upper limit of last class} - \text{Lower limit of 1st class} \quad \text{] grouped}$$

Coefficient of Range

$$\text{C.O.R} = \frac{X_{\max} - X_{\min}}{X_{\max} + X_{\min}}$$

answer will always lie b/w 0-1 or 0% - 100%.

Quartile Deviation

Also called "Semi Inter Quartile Range".

$$Q.D = \frac{Q_3 - Q_1}{2}$$

Coefficient of Quartile Deviation

$$\text{COQD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

It tells the variation % in the data.

Ex :	C.B	f	CF	Range = 120 - 26 = 94
	26 - 42	12	12	COR = $\frac{94}{146} = 0.64$
	42 - 56	18	30	
	56 - 78	30	60	$Q_1 = 56 + \frac{22}{30} (35 - 30) = 59.67$
	78 - 92	22	82	
	92 - 106	18	100	$Q_3 = 106 + \frac{14}{40} (105 - 100) = 107.75$
	106 - 120	40	140	

$$Q.D = \frac{107.75 - 59.67}{2} = 24.04$$

$$\text{COQD} = \frac{107.75 - 59.67}{107.75 + 59.67} = 28.7\% \text{ or } 0.28$$

Mean Deviation

The avg. of the absolute deviations taken from some central part of the data such as mode, median, mean.

$$MD_{\text{mean}} = \frac{\sum |X - \bar{X}|}{n} \quad \left. \vphantom{\frac{\sum |X - \bar{X}|}{n}} \right\} \text{ungrouped data}$$

where $\bar{X} = \frac{\sum X}{n}$, $\tilde{X} = \left(\frac{n+1}{2}\right)^{\text{th}}$, $\hat{X} = \text{most occurring value}$.

$$MD_{\text{median}} = \frac{\sum |X - \tilde{X}|}{n} \quad \left. \vphantom{\frac{\sum |X - \tilde{X}|}{n}} \right\} \text{ungrouped data}$$

$$MD_{\text{mode}} = \frac{\sum |X - \hat{X}|}{n} \quad \left. \vphantom{\frac{\sum |X - \hat{X}|}{n}} \right\} \text{ungrouped data}$$

$$MD_{\text{mean}} = \frac{\sum f |X - \bar{X}|}{\sum f}$$

$$MD_{\text{median}} = \frac{\sum f |X - \tilde{X}|}{\sum f}$$

$$MD_{\text{mode}} = \frac{\sum f |X - \hat{X}|}{\sum f}$$

} grouped data

NOTE: when not specified, only calculate MD_{mean} .

when question is that "prove that mean deviation from median is minimum" then calculate all three.

Lecture 11:

Coefficient of Mean Deviation

$$\text{Coefficient of } MD_{\text{mean}} = \frac{MD_{\text{mean}}}{\text{mean}}$$

$$COMD_{mode} = MD_{mode}$$

$$COMD_{median} = MD_{median}$$

Ex:	x	f	fx	$f x-\bar{x} $	MD _{mean}	$\bar{x} = \frac{2034}{130}$
	12	20	240	73		
	14	30	420	49.5		
	16	45	720	15.75		
	18	23	414	84.05		
	20	12	240	82.2		

$$= 130 = 2034 = 244.5$$

$$COMD_{mean} = \frac{1.88}{15.65}$$

$$= 0.1201$$

So, 12%

CB	CF	$f x-\bar{x} $	$f x-\hat{x} $
11-13	20	73.4	76.2
13-15	50	50.1	54.3
15-17	95	14.85	8.55
17-19	118	53.59	50.37
19-21	130	51.96	50.28

$$= 243.9 = 239.7$$

$$MD_{median} = \frac{\sum f|x-\tilde{x}|}{\sum f}$$

$$\tilde{x} = 15 + 2 \frac{(65-50)}{45}$$

$$\tilde{x} = 15.67$$

$$MD_{median} = \frac{243.9}{130}$$

$$MD_{median} = 1.876$$

$$COMD_{median} = \frac{1.876}{15.67}$$

$$MD_{mode} = \frac{\sum f|x-\hat{x}|}{\sum f}$$

$$\hat{x} = 15 + \frac{45-30}{45(45-23)} \times 2$$

$$= 15.81$$

$$MD_{mode} = \frac{239.7}{130}$$

$$MD_{mode} = 1.843$$

$$COMD_{mode} = \frac{1.843}{15.81} = 0.1165$$

So, 11.65%

$$= 0.1197$$

So, 11.97%

Variance

The avg of the square deviations taken from mean.

$$\sigma^2 = \frac{\sum (X - \bar{X})^2}{N}$$

(ungrouped)

↓
population

or

$$\sigma^2 = \frac{\sum X^2}{N} - \left(\frac{\sum X}{N} \right)^2$$

$$\text{(biased)} \quad S^2 = \frac{\sum (X - \bar{X})^2}{n}$$

(ungrouped)

↓
sample

$$\text{or} \quad S^2 = \frac{\sum (X - \bar{X})^2}{n-1}$$

(unbiased)

$$\sigma^2 = \frac{\sum f (X - \bar{X})^2}{\sum f}$$

(grouped)

↓
population

$$\text{or} \quad \sigma^2 = \frac{\sum f X^2}{\sum f} - \left(\frac{\sum f X}{\sum f} \right)^2$$

$$\text{(biased)} \quad S^2 = \frac{\sum n X^2}{n} - \left(\frac{\sum n X}{n} \right)^2$$

(grouped)

↓
sample

Properties

i) Variance of a constant is always zero.

$$V(C) = 0$$

$$\text{ii) } V(AX \pm B) = A^2 V(X)$$

where A and B are constants.

$$\text{iii) } V(AX \pm BY) = A^2 V(X) + B^2 V(Y)$$

X and Y are independent variables

$$\text{iv) } V(XY) = V(X) \cdot V(Y)$$

$$\text{v) } V\left(\frac{X}{A}\right) = \frac{1}{A^2} \cdot V(X)$$

Standard Deviation

the square root of variance.

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{N}}$$

$$\text{or } \sigma = \sqrt{\frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2}$$

ungrouped
population

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}}$$

$$\text{or } \sigma = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

grouped
population

$$S_{(\text{biased})} = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$S_{(\text{unbiased})} = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

ungrouped
sample

$$S = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

grouped
sample

Lecture 12:

Coefficient of Variance

Variation in 2 or more data sets ~~cannot~~ with different units of measurements cannot be compared unless we have relative measure of dispersion. To overcome this problem we have relative measure of dispersion which is known as coefficient of variance.

$$CV = \frac{\text{standard deviation}}{\text{mean}} \times 100.$$

$$SD / V \propto CV$$

$$CV \propto \frac{1}{\text{consistency}}$$

Ex: the data is about profit of two organizations during last 5 months.

X	Y
30	48
25	93
18	10
20	3
32	20

- (a) According to your opinion which org. is best w.r.t profit & why?
 (b) Which org. is showing more consistent performance w.r.t profit & why?

$$\bar{X} = \frac{\sum X}{n} = \frac{125}{5} = 25$$

$$SD_x = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2} = \sqrt{654.6 - 625} = 5.44$$

$$\text{Var}(X) = \frac{\sum X^2}{N} - \left(\frac{\sum X}{n}\right)^2 = 654.6 - 625 = 29.6$$

$$CV_x = \frac{5.44}{25} \times 100 = 21.76$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{174}{5} = 34.8$$

$$SD_y = \sqrt{\frac{\sum Y^2}{n} - \left(\frac{\sum Y}{n}\right)^2} = \sqrt{2292.4 - 1211.04} = 32.88$$

$$\text{Var}(Y) = \frac{\sum Y^2}{n} - \left(\frac{\sum Y}{n}\right)^2 = 2292.4 - 1211.04 = 1081.36$$

$$CV_y = \frac{32.88}{34.8} \times 100 = 94.48$$

(a) Best Profit means greater mean.

So, Company Y is best w.r.t profit

(b) more consistency means Lesser coefficient of Variance.

So, Company X is more consistent.

Probability

* measure of degree of belief in a statement

* quantitative measure uncertainty

Basic concepts :

(1) Experiment — a well defined activity that produces a result.

(2) Trial — single performance of experiment

(3) Outcomes — results obtained from an experiment.

(4) Sample Space — collection of all possible outcomes.
denoted by S or Ω

(5) Events & its Types — subset of sample space.
denoted by capital letters.

(6) Mutually Exclusive events

(7) Exhaustive Events

(8) Equally Likely Events — event that is as likely to occur as other events.

(9) Sure Events — event that consist of sample space itself event of sample must occur in a trial.

(10) Null Events / Impossible Events — contains no outcomes.
denoted by ϕ .

(11) Dependent Events

(12) Independent Events

Scientific Experiment

Experiment that gives identical outcomes when it is related under identical conditions.

Random Experiment

Experiment that gives different outcomes when it is repeated large no. of times under similar conditions.

Properties.

- i) Can be repeated any no. of times
- ii) All outcomes are known in advance.
- iii) Have more than one outcome.
- iv) Outcome can not be predicted in advance.

Lecture 13:

* $a^n \rightarrow a = \text{no. possible answers} ; n = \text{no. of events}$
gives all possible ways.

Event

• Simple Event

an event that contains only one sample point.

• Compound Event / Composite Event

an event that contains more than one sample points.

Mutually Exclusive Events

Two Events which cannot occur together i.e. $A \cap B = \emptyset$ are called non-overlapping events, disjoint events or mutually exclusive events.

Not-Mutually Exclusive Events

Two Events which can occur together i.e. $A \cap B \neq \emptyset$ are called overlapping, joint or not mutually exclusive events.

Exhaustive Events

Two or more events defined in same space. Their Union generates whole sample space.

Independent Events

if occurrence of one event does not effect the occurrence of another event.

Dependent Events

if occurrence of one event effects the occurrence of another event.

Counting Rules

• Rule of multiplication

The number of outcomes in the experiment is given by,

$$\prod_{i=1}^k n_i = n_1 \cdot n_2 \cdot n_3 \cdot \dots \cdot n_k$$

where k = no. of operations

n = no. of ways in which operations can be performed.

* terms used "and", " $\cap$ ".

Ex: In how many ways a three digit no. can formed using digits 1, 2, 3, 4, 5, 6

(a) if no. is even

(i) with repetition

$$\underline{6} \underline{6} \underline{3} = 108 \text{ ways}$$

(ii) without repetition

$$\underline{4} \underline{5} \underline{3} = 60 \text{ ways}$$

(b) if no. is > 300

(i) with repetition $\underline{4} \underline{6} \underline{6} = 144 \text{ ways}$

(ii) without repetition $\underline{4} \underline{5} \underline{4} = 80 \text{ ways}$

Ex : In how many ways 3 men 4 women can be arranged in a row

(a) without restriction

$$7 \ 6 \ 5 \ 4 \ 3 \ 2 \ 1 = 5040 \text{ ways}$$

(b) alternative (only possible if difference is one)

$$4 \ 3 \ 3 \ 2 \ 2 \ 1 \ 1 = 144 \text{ ways}$$

first fix place for the one who are more in quantity.

(c) men and women are together

$$4 \ 3 \ 2 \ 1 \ 3 \ 2 \ 1 = 144$$

$$3 \ 2 \ 1 \ 4 \ 3 \ 2 \ 1 = 144 +$$

288 ways.

(d) two people are always together



consider them 1

$$\text{So, } 6! \times 2! = 1440$$

no. of people who want to sit together

(e) two people refuse to sit together.

No. of ways people can sit - no. of ways two people sit together

$$= 5040 - 1440$$

$$= 3600$$

Lecture 14:

Ex: If a multiple choice test consists of 5 Qs.

each with four possible answers among which only 1 is correct.

(a) No. of ways a student can check off one answer to each question.

$$4^5 \text{ (4 options \& 5 Qs)} = 1024 \text{ ways.}$$

(b) No. of ways a student can mark off one answer to each question and get all answers wrong.
 $3^5 = 243$ ways.

Rule of Addition

number of outcomes is given by,

$$\sum_{i=1}^k n_i = n_1 + n_2 + n_3 + \dots + n_k$$

where k = no. of operations.

n = no. of ways operation can be performed.

Assume that 1 operation can be performed one time and no operation can be performed together.

Ex: In the medical study, patients are classified according to their blood group AB^+ , AB^- , A^+ , A^- , B^+ , B^- , O^+ , O^- along with their blood pressure condition low, high, normal.

(a) No. of ways patient can be classified according to blood group and bp type.

$$8 \times 3 = 24 \text{ ways}$$

$$\begin{matrix} \text{(no. of blood)} \\ \text{groups} \end{matrix} \times \begin{matrix} \text{(no. of types)} \\ \text{of bp} \end{matrix}$$

(b) No. of ways patient can be classified acc. to blood group or bp type.

$$8 + 3 = 11 \text{ ways.}$$

Permutation Type 1

ordered arrangement of " r " distinct objects.

No. of ways of ordering " n " objects are taken " r " at a time and is defined as,

$${}^n P_r = \frac{n!}{(n-r)!} \quad 0 \leq r \leq n$$

Ex: 3 lottery tickets for 1st, 2nd & 3rd prizes are drawn from group of 40 tickets.

(a) Find no. of sample points for avoiding the 3 prizes if each contestant holds 1 ticket.

$$n = 40, \quad r = 3$$

$${}^n P_r = {}^{40} P_3 = 59280 \text{ ways.}$$

Permutation Type 2

Permutation of n objects but ^{out} of n objects n_1 alike of 1st kind n_2 alike of 2nd kind and so on.

$${}^n P_{n_1, n_2, n_3, \dots, n_k} = \frac{n!}{n_1! \cdot n_2! \cdot n_3! \cdot \dots \cdot n_k!}$$

Ex: STATISTICIAN

$$S = 2 = n_1$$

$$C = 1 = n_5$$

$$T = 3 = n_2$$

$$N = 1 = n_6$$

$$A = 2 = n_3$$

$$I = 3 = n_4$$

$$n = 12 \text{ so, } \frac{12!}{2! \cdot 3! \cdot 2! \cdot 3! \cdot 1! \cdot 1!} = 3326400 \text{ ways}$$

Rule of Combination

arrangement of " r " objects. No importance of order.

$${}^n C_r = \frac{n!}{(n-r)! \cdot r!}$$

Ex: In how many different ways 11 men of football team can be selected from a squad of 20 people.

(a) If their positions are not ignored.

$${}^{20}P_{11} = 6.7044 \times 10^{12} \text{ ways.}$$

(b) If their positions are ignored.

$${}^{20}C_{11} = 167960 \text{ ways.}$$

Ex: From group of 4 men 5 women how many committees of ~~4~~ 3 people are possible?

(a) no restriction.

$$n = 4 + 5 = 9$$

$${}^9C_3 = 84 \text{ ways.}$$

(b) 1 men & two women.

$${}^4C_1 \times {}^5C_2 = 40 \text{ ways.}$$

(c) 2 men and 1 women if a specific man must be in the committee.

$${}^1C_1 \times {}^3C_1 \times {}^5C_1 = 15 \text{ ways.}$$

Ex: A shipment of 12 TV sets contains 3 defective sets. In how many ways a person can purchase 5 sets.

(a) all are non-defective

$$n = 12$$

$${}^9C_5 \times {}^3C_0 = 126 \text{ ways.}$$

(b) All are defective

No possibility.

(c) 2 ^{non-}defective sets

$${}^9C_2 \times {}^3C_3 = 36 \text{ ways.}$$

(d) 1 defective

$${}^9C_4 \times {}^3C_1 = 378 \text{ ways.}$$

(e) Atleast 1 defective

$$({}^3C_1)({}^9C_4) + ({}^3C_2)({}^9C_3) + ({}^3C_3)({}^9C_2) = 666 \text{ ways.}$$

Lecture 15 :

Probability

- * Objective Approach (same ans)
- * Subjective Approach (varies from person to person)

$$P(A) = \frac{\text{No. of favourable outcomes}}{\text{total outcome}}$$

Ex : From 4 red, 5 green & 6 yellow marbles. Nine are selected randomly.

- How many ways 9 marbles can be selected?
- In how many ways 2 red, 3 green and 4 yellow marbles can be selected?
- Calculate probability of part (b).
- Probability that 3 marbles of each colour are selected.
- 3 green and 4 yellow marbles are selected. Calculate probability.
- No. of ways 2 yellow marbles are selected.
- Probability of part (f).

$$4 + 5 + 6 = 15 \text{ marbles}$$

$$(a) {}^{15}C_9 = 5005 \text{ ways}$$

$$(b) ({}^4C_2)({}^5C_3)({}^6C_4) = 900 \text{ ways}$$

$$(c) \text{ Probability} = \frac{900}{5005} = 17.98\%$$

$$(d) \quad ({}^4C_3)({}^5C_3)({}^6C_3) = 800 \text{ ways}$$

$$\text{Probability} = \frac{800}{5005} = 15.98\%$$

$$(e) \quad ({}^5C_3)({}^6C_4)({}^7C_2) = 900 \text{ ways}$$

$$\text{Probability} = \frac{900}{5005} = 17.98\%$$

$$(f) \quad ({}^6C_2)({}^9C_7) = 540 \text{ ways}$$

$$(g) \quad \text{Probability} = \frac{540}{5005} = 10.78\%$$

Statistical / Relative Frequency Definition

$$P(A) = \lim_{n \rightarrow \infty} \frac{m}{n}$$

Axiomatic Definition

$$i) \quad 0 \leq P(A_i) \leq 1$$

$$ii) \quad P(A_1) + P(A_2) + \dots + P(A_n) = 1$$

iii) If A and B are mutually exclusive events then

$$\left[\begin{array}{l} P(A \cup B) = P(A) + P(B) \text{ such that} \\ P(A \cap B) = \phi \end{array} \right.$$

$$P(A \cap B) = \phi$$

→ addition rule for mutually exclusive

$$\left[\begin{array}{l} P(A \cup B) = P(A) + P(B) - P(A \cap B) \end{array} \right.$$

→ addition rule for non-mutually exclusive.

Ex : 10 files numbered 1 to 10. 1 file to be selected

(a) file no. is multiple of 5

$$A_1 = \{5, 10\}$$

$$P(A_1) = \frac{2}{10} = 0.2$$

(b) file no. is multiple of 3

$$A_2 = \{3, 6, 9\}$$

$$P(A_2) = \frac{3}{10} = 0.3$$

(c) file no. is multiple of 2

$$A_3 = \{2, 4, 6, 8, 10\}$$

$$P(A_3) = \frac{5}{10} = 0.5$$

(d) file no. is multiple of 2 or 5

$$A_4 = \{2, 4, 6, 8, 10, 5\}$$

$$P(A_3 \cap A_1) = \frac{1}{10} = 0.1 \quad // \text{ cuz } 10 \text{ is common}$$

Not mutually exclusive. So,

$$P(A_4) = P(A_3) + P(A_1) - P(A_3 \cap A_1)$$
$$= 0.5 + 0.2 - 0.1$$

$$P(A_4) = 0.6$$

(e) file no. is multiple of 3 or 5

$$A_5 = \{3, 6, 9, 5, 10\}$$

$$P(A_1 \cap A_2) = \phi$$

Mutually exclusive. So,

$$P(A_5) = P(A_1) + P(A_2)$$
$$= 0.2 + 0.3$$

$$P(A_5) = 0.5$$

Ex: 3 horses A, B, C. A is twice as likely to win as B, B is twice as likely to win as C. Probability that A or B wins.

$$P(C) = x$$

$$P(B) = 2x$$

$$P(A) = 4x$$

We know,

$$P(A) + P(B) + P(C) = 1$$

$$7x = 1$$

$$x = 1/7 = 0.142$$

$$\text{So, } P(A) = 0.571$$

$$P(B) = 0.285$$

$$\text{Now, Probability} = P(A) + P(B) = 0.571 + 0.285 = 0.795$$

Conditional Probability

Probability associated with reduced sample space.

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) \neq 0$$

Lecture 16:

Ex: A pair of die is rolled and no. are noted. Find the probability that the sum of no. is more than 8, given that no. on the second die is greater than the first.

	1	2	3	4	5	6
1	1+1	1+2	1+3	1+4	1+5	1+6
2	2+1	2+2	2+3	2+4	2+5	2+6
3	3+1	3+2	3+3	3+4	3+5	3+6
4	4+1	4+2	4+3	4+4	4+5	4+6
5	5+1	5+2	5+3	5+4	5+5	5+6
6	6+1	6+2	6+3	6+4	6+5	6+6

$$P(A) = \frac{10}{36} = 0.278$$

$$P(B) = \frac{15}{36} = 0.417$$

$$P(A \cap B) = \frac{4}{36} = 0.112$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{0.112}{0.417} = 0.268 \text{ or } 26.8\%$$

Ex: Probability that a regularly scheduled flight departs on time is $P(D) = 0.83$. and that it arrives on time is $P(A) = 0.92$. and that it departs and arrives in time is $P(D \cap A) = 0.78$. Find the probability that

(i) plane arrives in time given that it departed on time.

$$\text{So, } \frac{P(D \cap A)}{P(D)} = \frac{0.78}{0.83} = 0.939 / 93.9\%$$

(ii) plane departed on time given it arrived in time.

$$\text{So, } \frac{P(D \cap A)}{P(A)} = \frac{0.78}{0.92} = 0.847 / 84.7\%$$

Ex: Sample of 500 respondents was collected in a large metropolitan area. to study consumer behaviour with following results.

enjoying shopping		(m) male	female (f)
(e) yes		136	224
(n) No		104	36
		240	260

(a) Suppose selected respondent is female. Calculate probability that she doesn't enjoy shopping.

(b) Suppose selected respondent ~~selected~~ enjoyed shopping. Calculate probability that that person is male.

$$(a) \quad P\left(\frac{N}{F}\right) = \frac{P(N \cap F)}{P(F)} = \frac{0.072}{0.52} = 0.138$$

$$P(F) = \frac{260}{500} = 0.52$$

$$P(N \cap F) = \frac{30}{500} = 0.072$$

$$(b) \quad P\left(\frac{M}{E}\right) = \frac{P(M \cap E)}{P(E)} = \frac{0.272}{0.72} = 0.377$$

$$P(E) = \frac{360}{500} = 0.72$$

$$P(M \cap E) = \frac{136}{500} = 0.272$$

Ex: a vege basket contains 12 peppers in which 3 red, 4 green, 5 yellow. 3 peppers are selected at random.

(i) Probability are all are of different colors.

(ii) show that propability of selecting exactly two green pepper is $12/55$.

$$(i) \quad \frac{{}^3C_1 {}^4C_1 {}^5C_1}{{}^{12}C_3} = 0.272 \quad / \quad 27.2\%$$

$$(ii) \quad \frac{{}^4C_2 {}^8C_1}{{}^{12}C_3} = \frac{12}{55} = 0.218 \quad \text{or} \quad 21.8\%$$

Ex: Data about employment and gender in rural area is shown below.

	Unemployed (E')	Employed (E)	
M	206	412	$P(M) = \frac{618}{1281} = 0.482$
F	358	305	$P(E) = \frac{717}{1281} = 0.559$
	564	717	

$$P(M \cap E) = \frac{412}{1281} = 0.321$$

$$P\left(\frac{E}{M}\right) = \frac{P(E \cap M)}{P(M)} = \frac{0.321}{0.482} = 0.665$$

Probability that person is female or someone who is unemployed.

$$\begin{aligned}P(F \cup E') &= P(F) + P(E') - P(F \cap E') \\&= \frac{663}{1281} + \frac{564}{1281} - \frac{358}{1281} \\&= 0.678\end{aligned}$$

Are the two event M and E independent? No.

Independent Events:

$$\text{if } P(A \cap B) = P(A) \cdot P(B)$$

$$\text{So, } P\left(\frac{A}{B}\right) = P(A)$$

$$P\left(\frac{B}{A}\right) = P(B)$$

Dependent Events:

$$P(A \cap B) \neq P(A) \cdot P(B)$$

Lecture 17:

Ex: Let A be the event that machine can produce products of both types (good and bad). Let B be the event that machine can produce atmost 1 good product. If it is known that machine has to produce

(a) 3 products. then show A & B are independent.

(b) 4 products. then show A & B are dependent.

(possibilities)^{no. of products.}

(a) $2^3 = 8$ products.

	gg	gb	bg	bb
g	ggg	ggb	gbg	gbb
b	bgg	bgb	bbg	bbb

So,

$$A = \{ggb, gbg, gbb, bgg, bgb, bbg\}$$

$$P(A) = \frac{6}{8} = 0.75$$

$$B = \{gbb, bgb, bbg, bbb\}$$

$$P(B) = \frac{4}{8} = 0.5$$

$$P(A \cap B) = \frac{3}{8} = 0.375$$

$$P(A \cap B) = P(A) \cdot P(B)$$

$$0.375 = 0.375$$

Hence, A & B are independent.

$$(b) 2^4 = 16$$

	gg	gb	bg	bb
gg	gggg	gggb	ggbg	ggbb
gb	gbgg	gbgb	gbbg	gbbb
bg	bggg	bgbg	bgbg	bgbb
bb	bbgg	bbgb	bbbg	bbbb

So,

$$A = \{gggb, gggb, gggb, ggbg, ggbg, ggbg, gbbb, bggg, bggb, bggb, bgbb, bbgg, bbgg, bbgg, bbbg\}$$

$$P(A) = \frac{14}{16} = 0.875$$

$$B = \{gbbb, bgbb, bgbb, bgbb, bbbg, bbbg, bbbb\}$$

$$P(B) = \frac{5}{16} = 0.3125$$

$$P(A \cap B) = \frac{4}{16} = 0.25$$

$$P(A \cap B) = P(A) P(B)$$

$$0.25 \neq 0.27$$

So, Dependent Events.

Rules

$$i) P(A \cup B) = P(A) + P(B)$$

$$ii) P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

} for mutually exclusive

$$iii) P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$iv) P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

(v) $P(A) \rightarrow$ Probability of Event A taking place.

$P(\bar{A}) = 1 - P(A) \rightarrow$ probability of Event A not taking place.

Ex: A piece of equipment will function only when 3 components A, B & C are working. The probability of component A during 1 year is 0.15, of B is 0.05, of C is 0.10. What's the probability that equipment will fail to work before 1 year.

Events are independent. So,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) + P(A)P(B) + P(B)P(C) + P(A)P(C) + P(A)P(B)P(C)$$

$$= 0.15 + 0.05 + 0.10 + (0.15)(0.05) + (0.05)(0.10) +$$

$$(0.15)(0.10) + (0.15)(0.05)(0.10)$$

$$= \frac{11}{40} \text{ or } 0.275$$

Proportions (odds)

A : B

$$P(A) = \frac{A}{A+B}$$

$$P(B) = \frac{B}{A+B}$$

(Against B) (in favour of A)

(against A) (In favour of B)

Ex: The odds against student X solving a problem is 8 to 6. The odds in favour of student Y are 14 to 16.

(a) Probability that the problem will be solved if both work independently.

(b) Probability that non of them can solve the problem.

against X 8 to 6

$$P(X) = \frac{X}{X+Y} = \frac{6}{8+6} = 0.42$$

Favour Y 14 to 16

$$P(Y) = \frac{Y}{X+Y} = \frac{14}{14+16} = 0.46$$

$$P(X) = 6/14, \quad P(Y) = 14/30$$

$$\begin{aligned} P(X \cup Y) &= P(X) + P(Y) - P(X \cap Y) \\ &= 0.42 + 0.46 \end{aligned}$$

Lecture 18:

Ex: A husband and his wife appear in an interview for two vacancies of same post. Probability of husband's selection is $\frac{1}{7}$ and his wife's is $\frac{1}{5}$. What is the probability that both will be selected? Probability that at least one of them is selected? Probability that only husband will be selected? Probability that only wife will be selected? Probability that none of them are selected? Both are independent events.

$$P(H) = \frac{1}{7}, \quad P(W) = \frac{1}{5}$$

$$P(\bar{H}) = 1 - \frac{1}{7}, \quad P(\bar{W}) = 1 - \frac{1}{5}$$

$$= \frac{6}{7}$$

$$= \frac{4}{5}$$

$$\begin{aligned} (a) \quad P(H \cap W) &= P(H) * P(W) \\ &= \left(\frac{1}{7}\right) \left(\frac{1}{5}\right) = \frac{1}{35} \end{aligned}$$

$$\begin{aligned} (b) \quad P(H \cup W) &= P(H) + P(W) - P(H \cap W) \\ &= \left(\frac{1}{7}\right) + \left(\frac{1}{5}\right) - \frac{1}{35} = \frac{11}{35} \end{aligned}$$

$$\begin{aligned} (c) \quad P(H \cap \bar{W}) &= P(H) * P(\bar{W}) \\ &= \frac{1}{7} \left(\frac{4}{5}\right) = \frac{4}{35} \end{aligned}$$

$$\begin{aligned} (d) \quad P(\bar{H} \cap W) &= P(\bar{H}) * P(W) \\ &= \left(\frac{6}{7}\right) \left(\frac{1}{5}\right) = \frac{6}{35} \end{aligned}$$

Note : Mutually Exclusive events are always dependent.

$$\begin{aligned}P(\bar{H} \cap \bar{W}) &= P(\bar{H}) * P(\bar{W}) \\&= \left(\frac{6}{7}\right) \left(\frac{4}{5}\right) \\&= \frac{6}{5}\end{aligned}$$

Ex: $P(A) = 0.60$, $P(B) = 0.40$, $P(A \cap B) = 0.38$

find (i) $P\left(\frac{B}{A}\right)$ (ii) $P(A \cup B)$ (iii) relationship b/w A & B

$$(a) P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{0.38}{0.60} = 0.633$$

$$\begin{aligned}(b) P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\&= 0.60 + 0.40 - 0.38 \\&= 0.62\end{aligned}$$

(c) $P(A \cap B) \neq P(A) \cdot P(B)$ So, independent event
 $P(A \cap B) \neq 0$ So, not mutually exclusive event.

Multiplicative Law

$$P(A \cap B) = P(A) \cdot P\left(\frac{B}{A}\right)$$

$$P(B \cap A) = P(B) \cdot P\left(\frac{A}{B}\right)$$

$$P(A \cap B \cap C \cap D) = P(A) \cdot P\left(\frac{B}{A}\right) \cdot P\left(\frac{C}{A \cap B}\right) \cdot P\left(\frac{D}{A \cap B \cap C}\right)$$

Law of Total Probability

$$\begin{aligned}P(A) &= P(B \cap A) + P(C \cap A) \\&= P(B) \cdot P\left(\frac{A}{B}\right) + P(C) \cdot P\left(\frac{A}{C}\right)\end{aligned}$$

Ex: Box A contains 5 green & 7 red balls. Box B contains 3 green 3 red 6 yellow balls. A box is selected at random and a ball is drawn from it.
 (a) Probability that green ball is selected?

$$P(A) = \frac{1}{2}$$

$$P(B) = \frac{1}{2}$$

$$P\left(\frac{G}{A}\right) \quad P\left(\frac{G}{A}\right)'$$

$$P\left(\frac{G}{B}\right) \quad P\left(\frac{G}{B}\right)'$$

$$P\left(\frac{G}{A}\right) = \frac{{}^5C_1({}^7C_0)}{{}^{12}C_1} = \frac{5}{12}$$

$$P\left(\frac{G}{A}\right)' = 1 - \frac{5}{12} = \frac{7}{12}$$

$$P\left(\frac{G}{B}\right) = \frac{{}^3C_1({}^9C_0)}{{}^{12}C_1} = \frac{3}{12}$$

$$P\left(\frac{G}{B}\right)' = 1 - \frac{3}{12} = \frac{9}{12}$$

Let G be the event that green ball is selected
 It can be selected in the following ways:

$$\begin{aligned} P(G) &= P(A \cap G) + P(B \cap G) \\ &= P(A) \cdot P\left(\frac{G}{A}\right) + P(B) \cdot P\left(\frac{G}{B}\right) \end{aligned}$$

$$= \frac{1}{2} \left(\frac{5}{12} \right) + \frac{1}{2} \left(\frac{3}{12} \right)$$

$$= \frac{1}{3}$$

Ex : In a certain factory 3 machines M_1, M_2, M_3 are producing springs of same length. of their production M_1 produces 2%, M_2 produces 1% and M_3 produces 3% defective springs of total production. of the total production of springs, M_1 produces 35%, M_2 produces 25% and M_3 produces 40%. If one spring is selected at random and found to be defective. what is probability that it is produced by M_1, M_2 or M_3 ?

$$P(M_1) = 35\%$$

$$P(M_2) = 25\%$$

$$P(M_3) = 40\%$$

$$P\left(\frac{D}{M_1}\right)$$

$$P\left(\frac{D}{M_1}\right)'$$

$$P\left(\frac{D}{M_2}\right)$$

$$P\left(\frac{D}{M_2}\right)'$$

$$P\left(\frac{D}{M_3}\right)$$

$$P\left(\frac{D}{M_3}\right)'$$

$$= 0.02$$

$$= 0.98$$

$$= 0.01$$

$$= 0.99$$

$$= 0.03$$

$$= 0.97$$

$$P(D) = P(M_1 \cap D) + P(M_2 \cap D) + P(M_3 \cap D)$$

$$= P(M_1) P\left(\frac{D}{M_1}\right) + P(M_2) P\left(\frac{D}{M_2}\right) + P(M_3) P\left(\frac{D}{M_3}\right)$$

$$= 0.35(0.02) + 0.25(0.01) + (0.40)(0.03)$$

$$P(D) = 0.0215$$

$$P(M_1/D) = \frac{P(M_1 \cap D)}{P(D)} = \frac{P(M_1) \cdot P(D/M_1)}{P(D)} = \frac{0.35 \cdot 0.02}{0.0215}$$

$$= 0.3255$$

$$P(M_2/D) = \frac{P(M_2 \cap D)}{P(D)} = \frac{P(M_2) \cdot P(D/M_2)}{P(D)} = \frac{0.25 \cdot 0.01}{0.0215} = 0.116$$

$$P(M_3/D) = \frac{P(M_3 \cap D)}{P(D)} = \frac{P(M_3) \cdot P(D/M_3)}{P(D)} = \frac{0.40 \cdot 0.03}{0.0215} = 0.558$$

Lecture 19:

Base Theorem

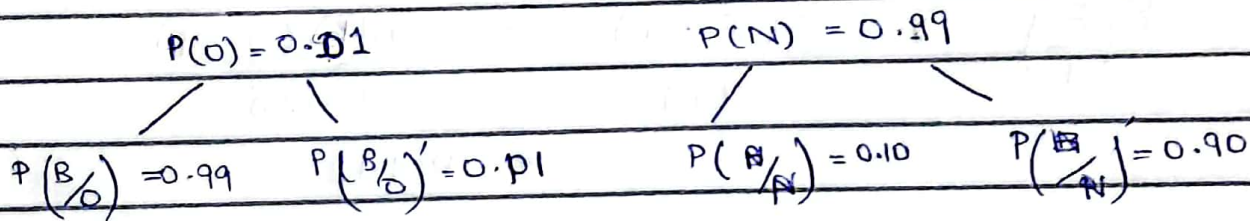
Probability of specific point divided by total probability.

e.g. $\frac{\text{Probability of defective in M1}}{\text{total Probability of defective.}}$

Ex: Suppose the probability that an electric circuit overloaded is 1%. The current passes through fuse that breaks with probability 99% when overloaded. when no more load occur, the probability that fuse does not break is

(a) Probability that fuse breaks.

(b) If fuse breaks then what is probability that it is due to overload.



$$\begin{aligned} \text{(a) } P(B) &= P(O \cap B) + P(N \cap B) \\ &= P(O) \cdot P\left(\frac{B}{O}\right) + P(N) \cdot P\left(\frac{B}{N}\right) = (0.01)(0.99) + (0.99)(0.10) \\ &= 0.1089 \end{aligned}$$

$$\begin{aligned} \text{(b) } P\left(\frac{O}{B}\right) &= \frac{P(O \cap B)}{P(B)} = \frac{P(O) * P\left(\frac{B}{O}\right)}{P(B)} \\ &= \frac{(0.01)(0.99)}{0.1089} = 0.09 \end{aligned}$$

Ex: In a certain uni 4% of male students and 1% of female student are taller than 6ft. 60% of students are male. If a student is selected randomly and is taller than 6ft.

(a) Probability that student is female.

$$\begin{array}{cc}
 P(M) = 0.60 & P(F) = 0.40 \\
 \swarrow \quad \searrow & \swarrow \quad \searrow \\
 P\left(\frac{T}{M}\right) & P\left(\frac{T}{M}\right)' \\
 = 0.04 & = 0.96 \\
 P\left(\frac{T}{F}\right) & P\left(\frac{T}{F}\right)' \\
 = 0.01 & = 0.99
 \end{array}$$

$$\begin{aligned}
 P(T) &= P(M \cap T) + P(F \cap T) \\
 &= P(M) P\left(\frac{T}{M}\right) + P(F) P\left(\frac{T}{F}\right) \\
 &= (0.60)(0.04) + (0.40)(0.01) \\
 &= 0.028
 \end{aligned}$$

$$\begin{aligned}
 P\left(\frac{F}{T}\right) &= \frac{P(F \cap T)}{P(T)} = \frac{(0.40)(0.01)}{0.028} \\
 &= 0.142
 \end{aligned}$$

Ex: The chances that Mr. X, Mr. Y & Mr. Z becoming a manager of a certain company are 4:2:3

The probability that the bonus scheme is launched and Mr. X becomes manager is 0.3, that of Mr. Y is 0.5 & Mr. Z is 0.8.

(a) Probability that bonus scheme will be launched

(b) Probability that Mr. Y is appointed as

Manager is scheme is launched.

$$P(X) = 4/9$$

$$P(Y) = 2/9$$

$$P(Z) = 1/3$$

$$P(B/X) = 0.3$$

$$P(B/Y) = 0.5$$

$$P(B/Z) = 0.8$$

(a) Let B be the event that bonus is introduced

$$P(B) = P(X \cap B) + P(Y \cap B) + P(Z \cap B)$$

$$= P(X) \cdot P\left(\frac{B}{X}\right) + P(Y) \cdot P\left(\frac{B}{Y}\right) + P(Z) \cdot P\left(\frac{B}{Z}\right)$$

$$= \left(\frac{4}{9}\right)(0.3) + \left(\frac{2}{9}\right)(0.5) + \left(\frac{1}{3}\right)(0.8)$$

$$= 0.510$$

$$(b) P\left(\frac{Y}{B}\right) = \frac{P(Y)P(B/Y)}{P(B)} = \frac{0.111}{0.510} = 0.217$$

Lecture 20 :

Random Variable

A real valued function, defined on sample spaces. It associates a numeric value to the each outcome of sample space/experiment. They are denoted by capital letters, and their respective values by small letters.

Note: Probability distribution / Discrete Probability Distribution / Probability mass Function means we are dealing with Discrete Data.

Probability Distribution

Arrangement of data into different groups or classes alongwith their probabilities.